Portia Wang

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Personal Website

Education

09/2024 – Present
Stanford University
Ph.D. in Communication (Media Psychology)
Advisor: Jeremy Bailenson

09/2022 – 04/2024
Stanford University
M.S. in Management Science and Engineering (Computational Social Science)
GPA: 4.13

09/2018 – 05/2022
Columbia University
B.S. in Computer Science, Minor in Applied Mathematics
Advisor: Steven Feiner
GPA: 4.01

Research Interests

Personalized Technology; Behavioral Modeling; Design Process; Virtual and Augmented Reality

Peer-Reviewed Publications

* denotes equal contributions

16. Tao, Y., Larrea, M., **Wang, P.**, Li, J., E., J, & Follmer, S. (To Appear). Reperceiving Everyday Sounds with Hidden Texture through Auditory Cueing. *The ACM Symposium on User Interface Software and Technology (UIST).*

15. Santoso, M., **Wang, P.**, Han, E., & Bailenson, J. N. (2026). Attitudinal, Social, and Behavioral Outcomes in Group Collaboration with Passthrough Mixed Reality. *Small Group Research.*

14. **Wang, P.**, Santoso, M., Han, E., & Bailenson, J. N. (2026). Synchrony and task engagement in immersive virtual environments: temporal dynamics, predictors, and psychological outcomes of collaborative behaviors. *Cognitive Science.*

13. Han, E., **Wang, P.**, Santoso, M., Rastogi, K. & Bailenson, J. N. (2026). Not Seeing Eye to Eye: The Effects of Perceptual Conflicts During Social Interactions in Mixed Reality. *Computers in Human Behavior.*

12. Bailenson, J., You, J., Markowitz, D., Petersen, G., Ratan, R., Santoso, M., & **Wang, P.** (2026). VHIL-E: An Expert LLM of the Virtual Human Interaction Lab's Research and Teaching. *Cyberpsychology, Behavior, and Social Networking.*

11. **Wang, P.**, Santoso, M., Han, E., Srirangarajan, T., & Bailenson, J. N. (2026). Virtual Placemaking: Self-built environments and revisiting shared memories in virtual reality increase group cohesion. *Journal of Computer-Mediated Communication.*

10. Santoso, M., **Wang, P.**, Han, E., & Bailenson, J. N. (2025). Virtual Reality Reduces Climate Indifference by Making Distant Locations Feel Psychologically Close. *Scientific Reports.*

9. Bailenson, J. N., DeVeaux, C., Han, E., Markowitz, D., Santoso, M., & **Wang, P.** (2025). Five Canonical Findings from 30 years of Psychological Experimentation in Virtual Reality. *Nature Human Behavior*.

8. Santoso, M., **Wang, P.**, Han, E., & Bailenson, J. N. (2025). Conversational Dynamics in Social Virtual Reality: A Large-Scale, Longitudinal Study of Speech Acts, Spatial Context, and Nonverbal Behavior. *Computers in Human Behavior*.

7. **Wang, P.**, Han, E., Queiroz, A. C., DeVeaux, C., & Bailenson, J. N. (2025). Predicting and Understanding Turn-Taking Behavior in Open-Ended Group Activities in Virtual Reality. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*.

6. Han, E., **Wang, P.**, DeVeaux, C., Harari, G., & Bailenson, J. N. (2024). Understanding the Role of Virtual Mobility on How and What People Create in Virtual Reality. *Thinking Skills and Creativity*.

5. **Wang, P.**, Miller, M. R., Queiroz, A. C., & Bailenson, J. N. (2024). Socially Late, Virtually Present: The Effects of Transforming Asynchronous Social Interactions in Virtual Reality. *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*.

4. Bailenson, J. N., Beams, B., Brown, J., DeVeaux, C., Han, E., Queiroz, A. C. M., Ratan, R., Santoso, M., Srirangarajan, T., Tao, Y., & **Wang, P.** (2024). Seeing the World Through Digital Prisms: Psychological Implications of Passthrough Video Usage in Mixed Reality. *Technology, Mind, and Behavior*.

3. **Wang, P.**, Miller, M. R., Han, E., DeVeaux, C., & Bailenson, J. N. (2024). Understanding virtual design behaviors: A large-scale analysis of the design process in Virtual Reality. *Design Studies*.

2. Aoyama, S.*, Liu, J. S.*, **Wang, P.**, Jain, S., Wang, X., Xu, J., … & Feiner, S. (2024). Asynchronously Assigning, Monitoring, and Managing Assembly Goals in Virtual Reality for High-Level Robot Teleoperation. *IEEE Conference Virtual Reality and 3D User Interfaces (VR)*.

1. Liu, J. S.*, **Wang, P.***, Tversky, B., & Feiner, S. (2022). Adaptive visual cues for guiding a bimanual unordered task in virtual reality. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*.

Book Chapters

3. **Wang, P.**, Srirangarajan, T., & Bailenson, J. N. (To appear). Social Processes of Learning in Virtual Reality. In Plass, J., Mayer, R. & Makransky, G. (Eds.), *The Handbook of Learning in Virtual Reality*. MIT Press.

2. Srirangarajan, T., **Wang, P.**, & Bailenson, J. N. (To appear). Multimodal Analytics in Virtual Reality. In Plass, J., Mayer, R. & Makransky, G. (Eds.), *The Handbook of Learning in Virtual Reality*. MIT Press.

1. **Wang, P.**, & Bailenson, J. N. (2026). Virtual reality as a research tool. In Reimer, T., van Swol, L. & Florack, A. (Eds.), *The Routledge Handbook of Communication and Social Cognition*. Routledge/Taylor and Francis.

Extended Abstracts and Posters

2. **Wang, P.**, Miller, M. R., & Bailenson, J. N. (2023). The Belated Guest: Exploring the Design Space for Transforming Asynchronous Social Interactions in Virtual Reality. *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*.

1. **Wang, P.**, Jain, S., Li, M., Aoyama, S., Wang, X., Song, S., ... & Feiner, S. (2023). Built to Order: A Virtual Reality Interface for Assigning High-Level Assembly Goals to Remote Robots. *Proceedings of the ACM Symposium on Spatial User Interaction.*

Works Under Review

9. Santoso, M., Petersen, G., **Wang, P.**, Chen, V., & Bailenson, J. N. (Under review). Testing the Boundaries of Construal Level Theory in Virtual Reality with AI Agent Interactions.

8. **Wang, P.**, Tao, Y., Santoso, M., & Bailenson, J. (Under review). Unpacking the process of watching a 360-degree video: How physiological responses and head rotation relate to affective and cognitive outcomes

7. Santoso, M., Markowitz, D., Petersen, G., Ratan, R., **Wang, P.**, You, J., & Bailenson, J. N. (Under review). AI Literacy Differentiates Longitudinal Style-Word Trajectories in Human-LLM Interaction.

6. Santoso, M., **Wang, P.**, & Bailenson, J. N. (Under review). The Relational Costs of Asymmetric Mixed Reality Headset Use in Large Groups.

5. DeVeaux, C., **Wang, P.**, Santoso, M., Han, E., & Bailenson, J. (Under review). Mind, Body, Avatar: The Role of Avatar Representation on Dancing in Social VR.

4. **Wang, P.**, Santoso, M., Tao, Y., & Bailenson, J. (Under review). Spatial configuration, movement, and group familiarity in co-watching 360-degree stories in social virtual reality.

3. Srirangarajan, T., Karmarkar, U., **Wang, P.**, & Bailenson, J. (Under review). Disentangling the roles of interactivity and decision framing on product valuation in immersive virtual reality.

2. Santoso, M., **Wang, P.**, McFall, L., & Bailenson, J. N. (Under review). Virtual Worlds as Behavioral Settings: Role Changes and Place Attachment Across Spatial Contexts.

1. Han, E., **Wang, P.**, Santoso, M., & Bailenson, J. N. (Under review). Mediated Mobility: The Effects of Embodied Travel in Virtual Reality.

Research Experience

10/2022 – Present
Graduate Research Assistant, Virtual Human Interaction Lab
Advised by Professor Jeremy Bailenson (Stanford University)
Conducts research on social behavior, presence, and emotion in virtual environments. Designs and evaluates proxemics, gaze, and mobility transformations to enhance immersive social interactions. Leads studies on modeling collaborative design, turn taking, and joint action in VR. Supports teaching in COMM166/266 Virtual People course and research mentorship.

07/2023 – 09/2023
Summer Research Fellow, Sensing, Interaction & Perception Lab
Advised by Professor Christian Holz (ETH Zürich)
One of 20 summer research fellows selected among 2,500+ applicants to conduct summer-long research in the Computer Science department at ETH Zürich. Conducted research on generating camera trajectories for VR and AR narratives.

09/2021 – 01/2023
Research Assistant, Computer Graphics and User Interfaces Lab
Advised by Professor Steven Feiner (Columbia University)
Conceptualized, built, and tested adaptive visual guidance systems in VR. Built immersive remote robot control systems for human-robot collaboration.

Awards and Honors

2024 **Stanford Graduate Fellowship in Science and Engineering**

2024 **Stanford McCoy Ethics Fellowship**

2023 **Stanford Sozo Graduate Fellowship**

2023 **Student Summer Research Fellowship, ETH Zürich**

2021 **Tau Beta Pi, Columbia University Chapter**

Teaching Experience

Fall 2025 **Teaching Assistant, Virtual People**
Instructor: Professor Jeremy Bailenson (Stanford University)

Fall 2024 **Teaching Assistant, Virtual People**
Instructor: Professor Jeremy Bailenson (Stanford University)

Spring 2022 **Head Teaching Assistant, 3D User Interface and Augmented Reality**
Instructor: Professor Steven Feiner (Columbia University)

Fall 2021 **Teaching Assistant, User Interface Design**
Instructor: Professor Brian Smith (Columbia University)

Talks and Presentations

Fall 2025 **Guest Lecture, Virtual People (Stanford University)**
Immersive Storytelling

Summer 2025 **Guest Lecture, Virtual Reality and Human Behavior (Stanford University)**
Immersive Storytelling

Fall 2024 **Guest Lecture, Virtual People (Stanford University)**
Generative AI and Virtual Reality

Professional Services

Conference Reviewer
ACM CHI
ACM CHI Late-Breaking Work
ACM CSCW
ACM DIS
ACM UIST
ACM VRST
IEEE ISMAR
IEEE VR
International Communication Association

Journal Reviewer
International Journal of Human-Computer Studies
Journal of Environmental Psychology
Multimedia Tools and Applications
Presence

Other Services **Columbia Application Development Initiative (ADI) Mentor (2021, 2022)**

Non-Research Industry Experience

Summer 2021 **Tencent (Product Management and Strategy Analyst Intern)** – Shenzhen, China
Composed strategy proposals for the smart city agenda based on AI Lab's 3D digital content capabilities; Delivered key strategy and roll-out plans for 3D digital content under the AI Lab; Defined objectives for the digital human enterprise solution.

Spring 2021 **Boston Consulting Group (Part-time Assistant)** – Remote
Engaged in a digital transformation case and an international digital strategy case in the TMT sector, conducted digital strategy research and expert calls and created digital strategy database construction and drafting of project case deliverables.

Summer 2019 **MoreVFX (Concept Design Intern)** – Beijing, China
Delivered and created early-stage virtual environments and concept arts for a science fiction movie production and facilitated the conceptualization and creation of commercial advertisements.

Skills

Research Design & Analysis
Survey design, User studies, Statistical hypothesis testing, Qualitative Interviews

Programming
Python, C#, R, Java, C++, C, Html, CSS

Hardware
HP Reverb G2 Omnicept, Meta Quest Series, Varjo XR-3, Microsoft Hololens Series, Pico

Software
Unity, 3DsMax, Photoshop, Figma, Mendeley, Git

Languages
English (fluent), Chinese (native)